

CHAPTER – 2

ENTITY ANALYSIS

2.1 INTRODUCTION

Entity Analysis is the process of identifying the entities, attributes, primary keys, foreign keys, and relationships required for the database system. It forms the foundation for designing an efficient and well-structured relational database.

For the ONE8 Footwear E-Commerce Database, the entities are identified to manage customers, products, categories, brands, shopping carts, wishlists, orders, payments, inventory, deliveries, and reviews.

2.2 ENTITY IDENTIFICATION

The following entities have been identified for the ONE8 Footwear E-Commerce Database. These entities represent the core components required to manage the online footwear shopping system.

2.2 ENTITY ATTRIBUTES

Entity	Description
Customer	Stores customer information.
Admin	Stores administrator details.
Product	Stores footwear product information.
Category	Stores footwear categories.
Brand	Stores brand information.
Shopping Cart	Stores products added to the customer's cart.
Wishlist	Stores customer's favourite products.
Order	Stores customer order details.
Order Item	Stores products included in each order.
Payment	Stores payment information.
Delivery	Stores shipment and delivery details.
Inventory	Stores available stock information.
Review	Stores customer reviews and ratings.
Offers	Stores discount offers and coupon details.

2.2.1 CUSTOMER TABLE

Attribute	Description
Customer_ID	Unique customer ID
Customer_Name	Customer name
Email	Email address
Phone_Number	Contact number
Password	Login password
Address	Customer address

2.2.2 ADMIN TABLE

Attribute	Description
Admin_ID	Unique administrator ID
Admin_Name	Administrator name
Email	Email address

Attribute	Description
Password	Login password

2.2.3 PRODUCT TABLE

Attribute	Description
Product_ID	Unique product ID
Product_Name	Product name
Price	Product price
Size	Shoe size
Color	Product color
Stock	Available stock
Category_ID	Category ID
Brand_ID	Brand ID

2.2.4 CATEGORY TABLE

Attribute	Description
Category_ID	Unique category ID
Category_Name	Category name

2.2.5 BRAND TABLE

Attribute	Description
Brand_ID	Unique brand ID
Brand_Name	Brand name

2.2.6 SHOPPING CART TABLE

Attribute	Description
Cart_ID	Unique cart ID
Customer_ID	Customer ID
Product_ID	Product ID
Quantity	Quantity selected

2.2.7 WISHLIST TABLE

Attribute	Description
Wishlist_ID	Unique wishlist ID
Customer_ID	Customer ID
Product_ID	Product ID

2.2.8 ORDER TABLE

Attribute	Description
Order_ID	Unique order ID
Customer_ID	Customer ID
Order_Date	Date of order
Total_Amount	Total amount
Order_Status	Order status

2.2.9 ORDER ITEM TABLE

Attribute	Description
Order_Item_ID	Unique order item ID
Order_ID	Order ID
Product_ID	Product ID
Quantity	Quantity ordered
Price	Product price

2.2.10 PAYMENT TABLE

Attribute	Description
Payment_ID	Unique payment ID
Order_ID	Order ID
Payment_Method	Payment method
Payment_Status	Payment status

2.2.11 DELIVERY TABLE

Attribute	Description
Delivery_ID	Unique delivery ID
Order_ID	Order ID
Delivery_Address	Delivery address
Delivery_Status	Delivery status

2.2.12 INVENTORY TABLE

Attribute	Description
Inventory_ID	Unique inventory ID
Product_ID	Product ID
Available_Stock	Available stock
Last_Updated	Last updated date

2.2.13 REVIEW TABLE

Attribute	Description
Review_ID	Unique review ID
Customer_ID	Customer ID
Product_ID	Product ID
Rating	Product rating
Review_Text	Customer review

2.2.14 OFFERS TABLE

Attribute	Description
Offer_ID	Unique offer ID
Coupon_Code	Discount coupon
Discount_Percentage	Discount percentage
Expiry_Date	Offer expiry date

2.3 PRIMARY KEYS

Entity	Primary Key
Customer	Customer_ID
Admin	Admin_ID
Product	Product_ID
Category	Category_ID
Brand	Brand_ID
Shopping Cart	Cart_ID
Wishlist	Wishlist_ID
Order	Order_ID
Order Item	Order_Item_ID
Payment	Payment_ID
Delivery	Delivery_ID
Inventory	Inventory_ID
Review	Review_ID
Offers	Offer_ID

2.4 FOREIGN KEYS

Table	Foreign Key	References
Product	Category_ID	Category
Product	Brand_ID	Brand
Shopping Cart	Customer_ID	Customer
Shopping Cart	Product_ID	Product
Wishlist	Customer_ID	Customer
Wishlist	Product_ID	Product
Order	Customer_ID	Customer
Order Item	Order_ID	Order
Order Item	Product_ID	Product
Payment	Order_ID	Order
Delivery	Order_ID	Order
Inventory	Product_ID	Product
Review	Customer_ID	Customer
Review	Product_ID	Product

2.5 RELATIONSHIP

Parent Entity	Relationship	Child Entity
Customer	One-to-Many	Order
Customer	One-to-Many	Shopping Cart
Customer	One-to-Many	Wishlist
Customer	One-to-Many	Review
Category	One-to-Many	Product
Brand	One-to-Many	Product
Order	One-to-Many	Order Item
Product	One-to-Many	Order Item
Order	One-to-One	Payment
Order	One-to-One	Delivery

Parent Entity	Relationship	Child Entity
Product	One-to-One	Inventory

2.5.1 RELATIONSHIP EXPLANATION

Relationship	Cardinality	Explanation
Customer → Order	One-to-Many (1:M)	One customer can place multiple orders.
Customer → Shopping Cart	One-to-Many (1:M)	One customer can add multiple products to the cart.
Customer → Wishlist	One-to-Many (1:M)	One customer can save multiple wishlist items.
Category → Product	One-to-Many (1:M)	One category contains multiple products.
Brand → Product	One-to-Many (1:M)	One brand offers multiple products.
Order → Order Item	One-to-Many (1:M)	One order contains multiple products.
Product → Order Item	One-to-Many (1:M)	One product can appear in many orders.
Order → Payment	One-to-One (1:1)	Every order has one payment record.
Order → Delivery	One-to-One (1:1)	Every order has one delivery record.
Product → Inventory	One-to-One (1:1)	Every product has one inventory record.

2.6 CONCLUSION

The identified entities, attributes, primary keys, foreign keys, and relationships provide the foundation for designing the ONE8 Footwear E-Commerce Database. These components help create an efficient relational database that supports customer management, product management, order processing, payment handling, inventory management, and delivery tracking. The entity analysis will be used to develop the ER Diagram and the relational database schema in the next phase of the project.